

STATISTICAL FIELD THEORY

The Ising Model

Consider a 2-dimensional lattice with N sites $i = \{1, \dots, N\}$, each of which is one of two states $s_i = \pm 1$ (spin up/spin down). This collection has energy

$$E = -B \sum_i s_i - J \sum_{\langle i,j \rangle} s_i s_j$$

if $J > 0$, neighbouring states prefer to be aligned $\rightarrow$ ferromagnet

external magnetic field neighbouring i,j

Recall that in the canonical ensemble, probability that i is in state s_i is

$$p[s_i] = \frac{e^{-\beta E[s_i]}}{Z}$$

where $k_B = 1$, $\beta = \frac{1}{T}$, and the partition function

$$Z(T, J, B) = \sum_{\{s_i\}} e^{-\beta E[s_i]}$$

Recall the free energy

$$F_{\text{thermo}}(T, B) = \langle E \rangle - TS = -T \log Z$$

We call the average spin of the configuration

$$m = \frac{1}{N} \langle \sum_i s_i \rangle \in [-1, 1]$$

the magnetisation. Note that

$$m = \frac{1}{N} \sum_{\{s_i\}} \frac{e^{-\beta E[s_i]}}{Z} \sum_i s_i = \frac{1}{N\beta} \frac{\partial \log Z}{\partial B}$$

Let's write

$$Z = \sum_m \sum_{\{s_i\} | m} e^{-\beta E[s_i]} = \sum_m e^{-\beta F(m)}$$

can depend on T , and hence on β
Hard to calculate

sum over $\{s_i\}$
s.t. $\frac{1}{N} \sum s_i = m$

in the limit as $N \rightarrow \infty$

$$= \frac{N}{Z} \int_{-1}^{+1} dm e^{-\beta F(m)}$$

We call $F(m)$ the effective free energy. Let

$$f(m) = \frac{F(m)}{N}$$

so
$$Z = \frac{N}{Z} \int_{-1}^{+1} dm e^{-\beta N f(m)}$$

We assume that $N \sim 10^{23} \gg 0$ while $\beta N \sim 1$, so this integral is well approximated by the value of the integrand at $m = m_{\text{min}}$, minimising f , i.e.

$$\left. \frac{\partial f}{\partial m} \right|_{m=m_0} = 0$$

Hence,
$$Z \approx e^{-\beta N f(m_{\text{min}})}$$

and
$$F_{\text{thermo}} \approx F(m_{\text{min}})$$

We will try to find $F(m) = N f(m)$ using mean field theory. Some of this is not going to work! To do this, we want to sum over $\{s_i\}$ with $\sum s_i = Nm$.

We first just replace each s_i with its mean value m

$$E = -B \sum_i s_i - J \sum_{\langle i,j \rangle} s_i s_j \Rightarrow \frac{E}{N} \approx -Bm - \frac{1}{2} J q m^2$$

neighbours

We have
$$m = \frac{N_{\uparrow} - N_{\downarrow}}{N} = \frac{2N_{\uparrow} - N}{N}$$

with the number of such configurations

$$\Omega = \frac{N!}{N_{\uparrow}! (N - N_{\uparrow})!}$$

By Stirling,

$$\frac{\log \Omega}{N} \approx \log 2 - \frac{1}{2} (1+m) \log(1+m) - \frac{1}{2} (1-m) \log(1-m)$$

With the mean field approximation,

$$e^{-\beta N f(m)} \approx \Omega(m) e^{-\beta E(m)}$$

and hence

$$f(m) \approx -Bm - \frac{1}{2} J q m^2 - T \left(\log 2 - \frac{1}{2} (1+m) \log(1+m) - \frac{1}{2} (1-m) \log(1-m) \right)$$

(calculating the minimum of $f(m)$, we get

$$m = \tanh(\beta B + \beta J q m)$$

the self-consistency equation. This can also be derived by assuming that each spin experiences an effective magnetic field $B_{\text{eff}} = B + J q m$, the mean field, thus replacing the tricky interactions.

Landau Approach to Phase Transitions

Recall that

$$f(m) \approx -Bm - \frac{1}{2}Jqm^2 - T\left(\log 2 - \frac{1}{2}(1+m)\log(1+m) - \frac{1}{2}(1-m)\log(1-m)\right).$$

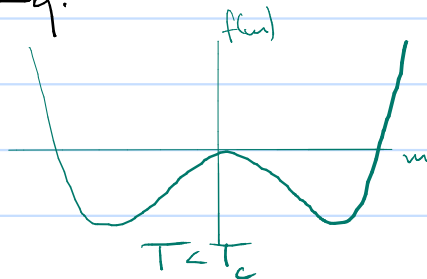
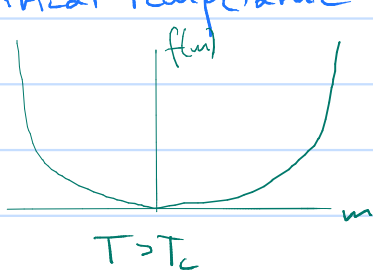
For small m ,

$$f(m) \approx -T \log 2 - Bm + \frac{1}{2}(T - Jq)m^2 + \frac{1}{12}Tm^4 + \dots$$

Let now $B=0$. We are interested in the locations of the minima of f , so we can drop the constants. Then

$$f(m) \approx \frac{1}{2}(T - Jq)m^2 + \frac{1}{12}Tm^4 + \dots$$

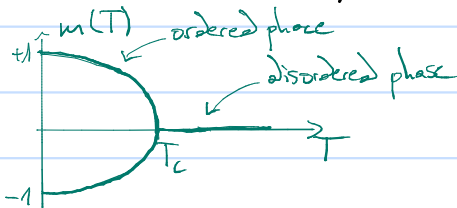
Let the **critical temperature** be $T_c = Jq$.



We see that at high energies, the magnetisation vanishes at the minimum $m=0$. At energies below T_c , the free energy is minimised at $m = \pm m_0$, where

$$m_0 \approx \sqrt{\frac{3(T_c - T)}{T}}$$

when $T_c \approx T$.



This kind of phase transition is called **continuous**, or **second order phase transition**.

Recall the heat capacity $C = \frac{\partial \langle E \rangle}{\partial T}$. In the canonical ensemble, $\langle E \rangle = -\frac{\partial \log Z}{\partial \beta}$, so

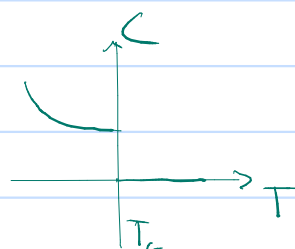
$$C = \beta^2 \frac{\partial^2}{\partial \beta^2} \log Z.$$

still neglecting the constant term

We still need to compute $\log Z$, i.e. compute $f(m_{\min})$. For $T > T_c$, $m_{\min} = 0 = f(m_{\min})$. For $T < T_c$, $f(m_{\min}) = -\frac{3}{4}(T_c - T)^2/T$.

We get

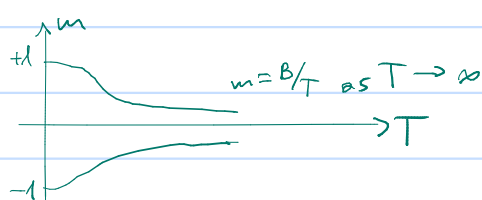
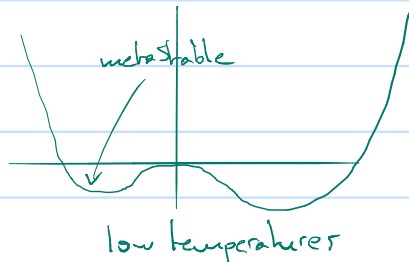
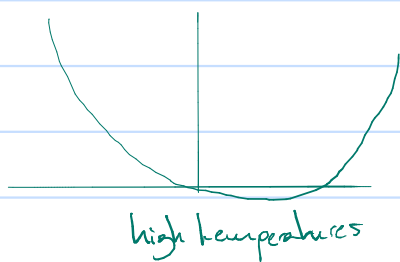
$$C = \frac{C}{N} = \begin{cases} 0 & \text{as } T \rightarrow T_c^+ \\ \frac{3}{2} & \text{as } T \rightarrow T_c^- \end{cases},$$



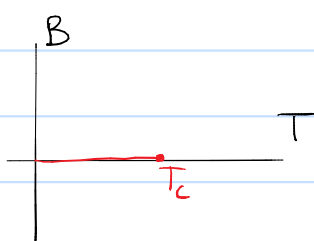
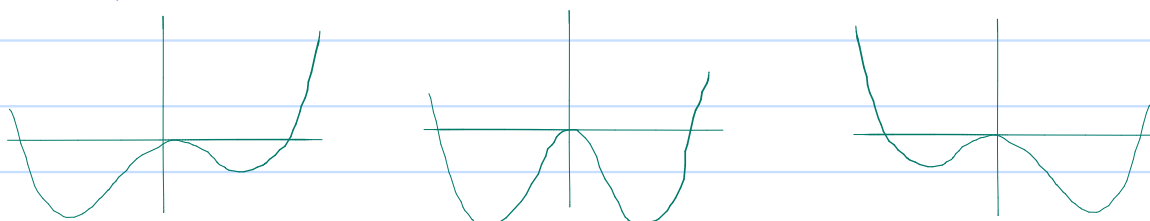
a discontinuous jump.

Let now $B \neq 0$. Then, ignoring constants,

$$f(m) \approx -Bm + \frac{1}{2}(T - Jq)m^2 + \frac{1}{12}Tm^4 + \dots$$



Here, there is no phase transition. Even the branch of m is determined by the sign of B . If we fix $T < T_c$ and vary B from negative to positive, we do get a **first order phase transition**.



Phase diagram:

Moving through the red line yields a 1st order phase transition, at T_c a 2nd order PT, after none $\rightarrow (B, T) = (0, T_c)$ is a critical point